Truong Vu-Quang

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Personal Statement

I'm a first-year master's student (M1) in the DataAI PhD Track at Institut Polytechnique de Paris and a former AI Resident at VinAI Research. My research spans computer vision and medical image analysis, with a particular interest in how detection and segmentation models can adapt to new settings with limited labeled data.

Education

Institut Polytechnique de Paris — Palaiseau, France
Master's in Data and Artificial Intelligence (M1). — 2026 - Present

Hanoi University of Science and Technology (HUST) — Hanoi, Vietnam
Bachelor in Computer Science, Talented Program. — 2019 - 2023

- **CGPA:** 3.90/4.0. Graduated #1 in the program.
- **Thesis:** SPARTA-GEMSTONE: A two-phase approach for efficient node placement in 3D wireless sensor networks under Q-Coverage and Q-Connectivity constraints.

Experience

Mohamed bin Zayed University of Artificial Intelligence (MBZUAI) — Remote
Visiting Researcher - Supervisor: Assist. Prof. Yutong Xie — 11/2025 - Now

- Conducted research in Medical AI under the supervision of Assist. Prof. Yutong Xie.

Qualcomm AI Research — Hanoi, Vietnam
AI Resident - Supervisor: Prof. Minh Hoai Nguyen — 04/2025 - 06/2025

- Continued the work at VinAI on low-rank prompt adaptation for open-vocabulary object detection, resulting in one paper accepted at ICCV 2025 Workshop.

VinAI Research — Hanoi, Vietnam
AI Resident - Supervisor: Prof. Minh Hoai Nguyen — 02/2023 - 03/2025

- **Efficiency-preserving Scene-adaptive Object Detection:** Developed a framework for multi-camera video streams, achieving state-of-the-art precision-efficiency trade-offs (4.65 mAP improvement). Accepted for oral presentation at BMVC 2024.
- **Low-Rank Prompt Adaptation for Open-Vocabulary Object Detection:** Pioneered a framework that optimizes textual prompts, achieving state-of-the-art performance across five datasets with minimal annotation requirements.
- **Unidentified License Plate Detection:** Built a detection pipeline for obscured license plates, integrating OCR, and was awarded Best Applied Rotation Project in Q2 2024.

MSO Lab, HUST — Hanoi, Vietnam
Member - Supervisor: Assoc. Prof. Thi Thanh Binh Huynh — 10/2020 - Present

- Conducted research on target and area coverage problems in wireless sensor networks (WSNs) and mentored new lab members. Three papers accepted in *Journal of Network and Computer Applications* (Q1, IF 7.7).

FPT Software — Hanoi, Vietnam
AI Engineer Intern — 08/2022 - 02/2023

- Developed a chatbot using Rasa to assist the IT Helpdesk team, automating user inquiries and improving support response times.

Publications

Conference Proceedings

1. **Truong Vu**, Minh Khoi Ho, and Yutong Xie, "Attention-Based Prototype Calibration for Multi-Rater Few-Shot Medical Image Segmentation," in *MICCAI 2026*.
2. Z. Zhang*, **V. Q. Truong***, and M. Hoai, "Low-Rank Prompt Adaptation for Open-Vocabulary Object Detection," in *ICCV 2025 Workshop*.
3. Z. Zhang, **V. Q. Truong**, and M. Hoai, "Efficiency-preserving scene-adaptive object detection," in *35th British Machine Vision Conference 2024*.

Journal Articles

1. **Truong Q. Vu**, Minh T. Trinh, Hien Q. Ta, Hanh T. Nguyen, and Binh T T. Huynh, "STEAM: Securing the connections in wireless sensor networks with varying target priority," *Computer Networks*, vol. 277, p. 112024, 2026.
2. **Q. T. Vu**, T. M. Trinh, T. H. Nguyen, V. C. Trinh, T. T. B. Huynh, X. T. Nguyen, and C. P. Huynh, "SPARTA-GEMSTONE: A two-phase approach for efficient node placement in 3D WSNs under Q-Coverage and Q-Connectivity constraints," *Journal of Network and Computer Applications*, vol. 239, p. 104175, 2025.
3. **Q. T. Vu**, P. T. Nguyen, T. H. Nguyen, T. T. B. Huynh, V. C. Trinh, and M. Gidlund, "Striking the perfect balance: Multi-objective optimization for minimizing deployment cost and maximizing coverage with harmony search," *Journal of Network and Computer Applications*, vol. 232, p. 104006, 2024.
4. N. T. Hanh, H. T. T. Binh, **V. Q. Truong**, N. P. Tan, and H. C. Phap, "Node placement optimization under q-coverage and q-connectivity constraints in wireless sensor networks," *Journal of Network and Computer Applications*, vol. 212, p. 103578, 2023.

Awards and Achievements

Best Applied Rotation Project of Quarter 2, VinAI Research 2024
Third prize, AI Hackathon: Face Analysis Challenge, Pixta Vietnam 2024
2 times **Sumitomo Sponsor Scholarship**, HUST 2023, 2020
Best Presentation Award for the graduation thesis in the Optimization committee, HUST 2023
First prize, SoICT Hackathon, Tiki Track 2023
Consolation Prize, SoICT Student Research Conference 2022
3 times **Study Encouragement Scholarship**, HUST 2022, 2021
Top 6 in National High School Exam with Math, Physics, and English combination 2019

Certifications

IELTS Academic: 8.0 (Reading 9.0, Listening 8.5, Writing 7.0, Speaking 6.5), British Council 2024

Skills

Technical: Java, Python, C/C++, SQL, LaTeX, MySQL, PostgreSQL, SQLite
Language: Proficient in English (reading, writing, and speaking)
AI Expertise: Computer Vision
Others: Academic research, teamwork, presentation skills

References

1. **Minh Hoai Nguyen**, *Professor* The University of Adelaide, Australia
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2. **Yutong Xie**, *Assistant Professor* Mohamed bin Zayed University of Artificial Intelligence
Email: yutong.xie@mbzuai.ac.ae
3. **Thi Thanh Binh Huynh**, *Professor* Hanoi University of Science and Technology, Vietnam
Email: binhht@soict.hust.edu.vn

*Equal contribution